nuclear phenomena themselves.

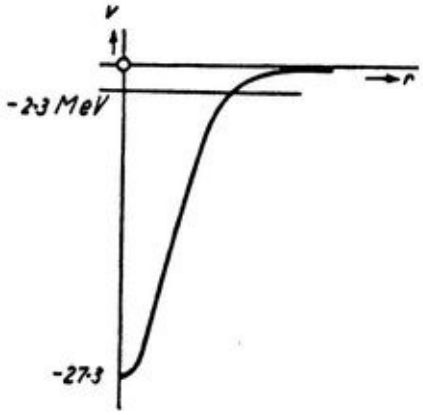


Figure 15.—Potential of the force between neutron and proton.

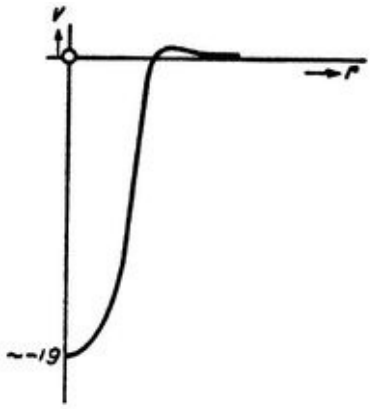


Figure 16.—Potential of the force between proton and proton.

This has already supplied us with a certain general view of the nature of the force operative between a proton and a neutron. But what about the force between two protons?

One might surmise, to start with, that electric repulsion only is operative between them, since the force between protons and neutrons would be sufficient in itself to explain both nuclear cohesion and the fact that nuclear matter in a stable state consists of an approximately equal number of protons and neutrons. For if some force were operative between protons and neutrons only, the symmetry would already be guaranteed, to begin with. But experience concerning the deflection of protons by protons proves that forces of attraction are acting between particles of the same kind—in other words, not only between protons but between neutrons, too—which forces of attraction are approximately equal to those acting between protons and neutrons. In the case of two protons the situation is more complicated, because the electric force of repulsion is superimposed on the nuclear force of attraction. But when dealing with very short distances, the force of repulsion